Generative AI for Video Generation (1)

COMP4431 Lab 4

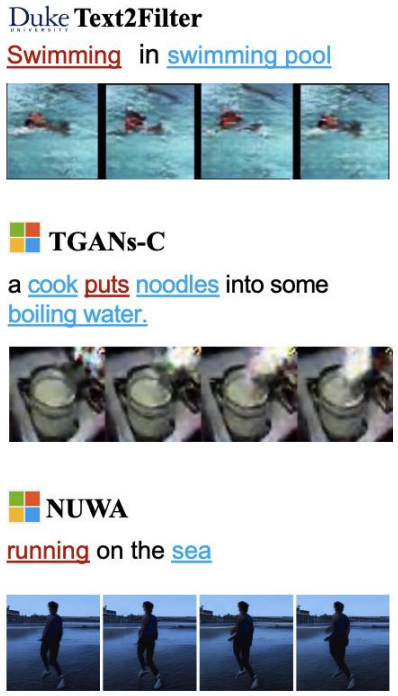
2026 update

Lab 04 Agenda

- Video Generation Models
  - Introductions of Video Generation Models
  - Products of Video Generation Models
  - Access to Video Generation Models
- Basic Video Generation Techniques
  - Text-to-video Generation
  - Image-to-video Generation
  - Video Editing
- Other Tools
  - Lip Sync Video
  - FramePack: Image-to-video Running Locally

Introductions of Video Generation Models

- Concept
  - Video generation models deal with the creation of synthetic videos.
    - New video clips are generated by describing the content of the desired video footage (text-to-video)
    - or applying the style and composition via text or image prompt to a source video (video-to-video).
- History
  - Early works are limited to simple scenes (low-resolution, single-object, and short-duration).
  - Benefiting from the success achieved by the diffusion model in the generative area, current works are generating more complex videos.





Zhou P, Wang L, Liu Z, et al. A survey on generative ai and llm for video generation, understanding, and streaming[J]. arXiv preprint arXiv:2404.16038, 2024.

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Products of Video Generation Models

- Video Generation Models in 2026
  - **Google's Gemini Omni**
    - On May 19th, 2026, Google introduces Gemini Omni, a model that can "create anything from any input".
    - Gemini Omni Flash generates videos from any mix of text, images, video and audio, and lets you edit them through conversation.
  - **ByteDance's Seedance 2.5**
    - On July 31st, 2026, ByteDance Seed introduces Seedance 2.5, a video creation model that generates video and audio together.
    - Seedance 2.5 can generate videos up to 30 seconds long in a single pass, using up to 30 images, 10 video clips and 10 audio clips as references.
  - Note: OpenAI's Sora (2024) was discontinued in 2026.



https://deepmind.google/models/gemini-omni/



https://seed.bytedance.com/seedance2_5

Products of Video Generation Models

- Available Generative Artificial Intelligence Products for Video Generation (Oct 2026)

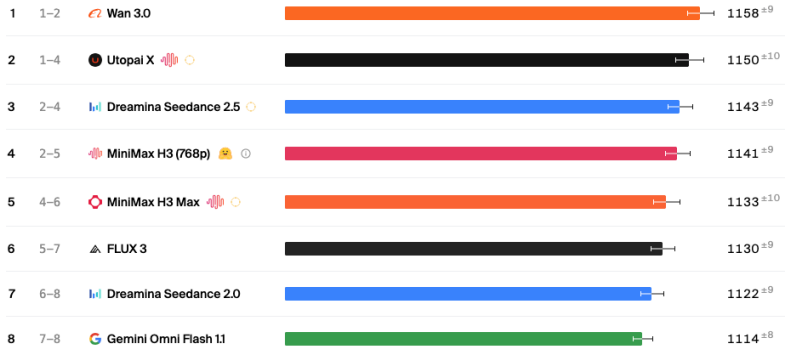
Company	Latest Model	Generation Type	Max Length	Max Resolution	Native Audio?	Other Features	Format
Runway	Gen-4.5 / Aleph 2.0	Text-to-video, image-to-video, video-to-video	10 sec	720p (4K upscale)	No	Aleph 2.0 video editing, 4K upscale	Website, API
Pika	Pika 2.5	Text-to-video, image-to-video	10 sec	1080p	No	Keyframes (2–5 frames), add/swap objects	Website, App, API
Google Veo	Veo 3.1	Text-to-video, image-to-video	8 sec	**4K**	Yes	First/last frame, extend to 148 sec	Website, API
Google Gemini	Gemini Omni Flash 1.1	Text-to-video, image-to-video, video-to-video	10 sec	4K (upscale)	Yes	Conversational editing, extend to 40 sec	Website, App, API
Kling AI	Kling 3.0	Text-to-video, image-to-video, video-to-video	15 sec	**4K**	Yes	Multi-shot (up to 6), multilingual lip sync	Website, App, API
ByteDance Seed	Seedance 2.5	Text-to-video, image-to-video, video-to-video	**30 sec**	1080p	Yes	Up to 50 references, timestamp editing	Website, App, API
Alibaba Wan	Wan 3.0	Text-to-video, image-to-video, video-to-video	**30 sec**	1080p	Yes	Up to 20 references (incl. documents)	API
MiniMax Hailuo	MiniMax H3	Text-to-video, image-to-video, video-to-video	15 sec	2K	Yes	Dialogue in 11 languages	Website, API, Open weights
xAI Grok Imagine	Grok Imagine Video 1.5	Text-to-video, image-to-video	15 sec	1080p	Yes	Lip-synced speech, 14 reference images	Website, App, API
Luma AI	Ray3.2	Text-to-video, image-to-video, video-to-video	10 sec	1080p (4K upscale)	No	16 keyframes, facial performance transfer	Website, API
Vidu	Vidu Q4 Preview	Image-to-video, reference-to-video	16 sec	**4K**	Yes	15 reference images, voice cloning	Website, API
PixVerse	PixVerse V6	Text-to-video, image-to-video, reference-to-video	15 sec	1080p	Yes	Multi-shot, first/last frame, lip sync	Website, App, API
LTX (Lightricks)	LTX-2.5	Text-to-video, image-to-video, video-to-video	20 sec	**4K**	Yes	Multi-shot; runs on a 16 GB GPU	Website, API, Open weights
Tencent Hunyuan	HunyuanVideo 1.5	Text-to-video, image-to-video	10 sec	720p (1080p upscale)	No	8.3B params; runs on a 14 GB GPU	App, API, Open weights
Midjourney	V1 Video	Image-to-video	5 sec	720p	No	Extend to 21 sec, loop	Website

Sources: official product pages and docs of each company, checked Oct 13, 2026

Products of Video Generation Models

- Artificial Analysis Video Arena ranks video models by Elo, based on blind human votes between two videos made from the same prompt.
  - Wan 3.0 (Alibaba, Aug 2026) ranks 1st in text-to-video and video editing.
  - MiniMax H3 Max (fal, Aug 2026; built on MiniMax's open-source H3) leads image-to-video.

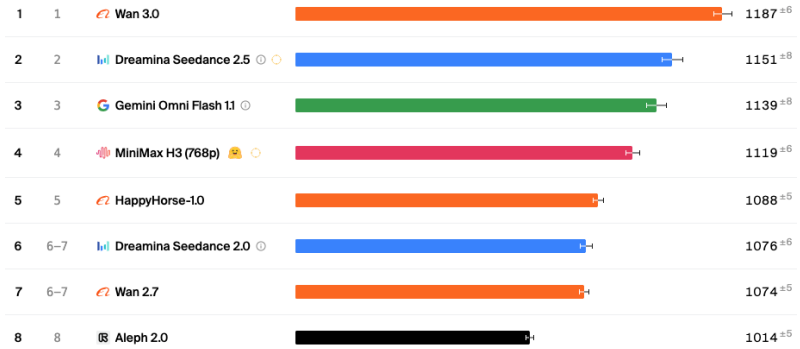
Leaderboard data as of Oct 13, 2026



- Text-to-Video
  - generates a video, together with its audio, from a text prompt alone.



- Image-to-Video
  - animates a given source image into a video; the image fixes the subject and composition.



- Video Editing
  - edits an existing video from a text instruction while keeping its original audio.

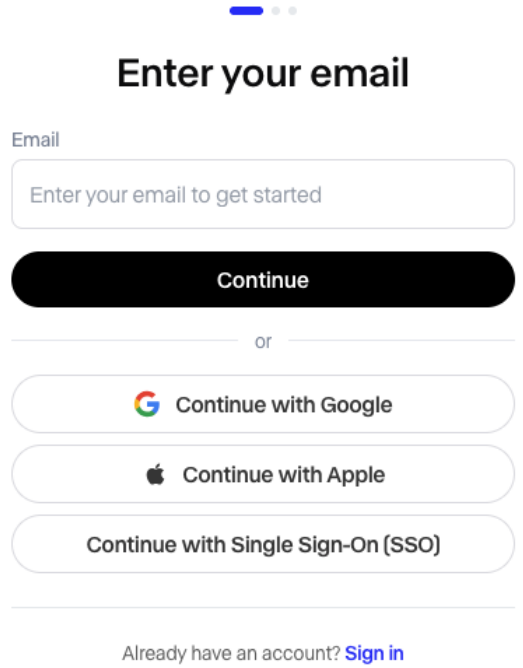
https://artificialanalysis.ai/video/leaderboard/text-to-video
https://artificialanalysis.ai/video/leaderboard/image-to-video
https://artificialanalysis.ai/video/leaderboard/video-editing

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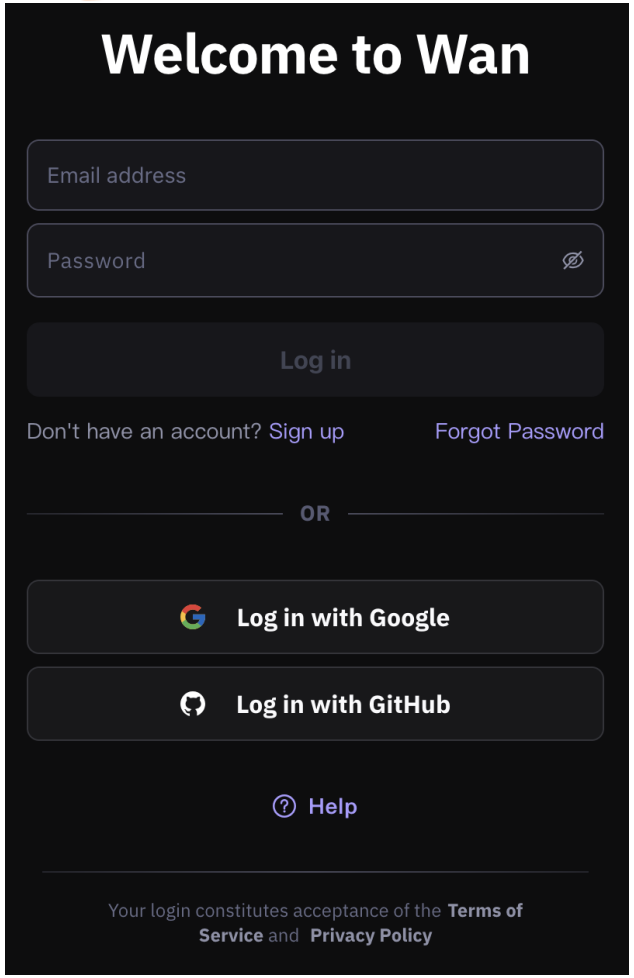
Access to Video Generation Models (1)

- Runway ML
  - Navigate to https://app.runwayml.com/
  - Log in using your account and password
    - You can create account with different email address.
    - You can login with Google account.
    - You can login with Apple account.
  - Select a plan
    - Free Plan ($0): Can't remove watermarks or upscale to 4K; only a selection of models; 125 one-time credits (125 credits = 10s of Gen-4.5, 20s of Gen-4 Turbo)
    - Standard Plan ($12): Remove watermarks; access to all video and image models. 625 credits/month (625 credits = 52s of Gen-4.5, 104s of Gen-4 Turbo)
    - Pro Plan ($28): 2250 credits/month (2250 credits = 187s of Gen-4.5, 375s of Gen-4 Turbo)
  - Runway also hosts third-party models: Seedance 2.5, Veo 3.1, Wan 3.0, MiniMax H3
  - Prompt to create a video



Access to Video Generation Models (2)

- Wan
  - Navigate to https://create.wan.video
  - Log in using your account and password
    - You can create account with different email address.
    - You can login with Google account.
    - You can login with GitHub account.
  - Select a plan
    - Free Plan ($0): Can't remove watermarks or generate 1080p; 6 credits per day by daily check-in (6 credits = 2 s of Wan 3.0 at 720p; unused credits expire after 24 hours).
    - Pro Plan ($5): Remove watermarks, 1080p, 300 credits/month
    - Premium Plan ($20): 1200 credits/month
  - Prompt to create a video

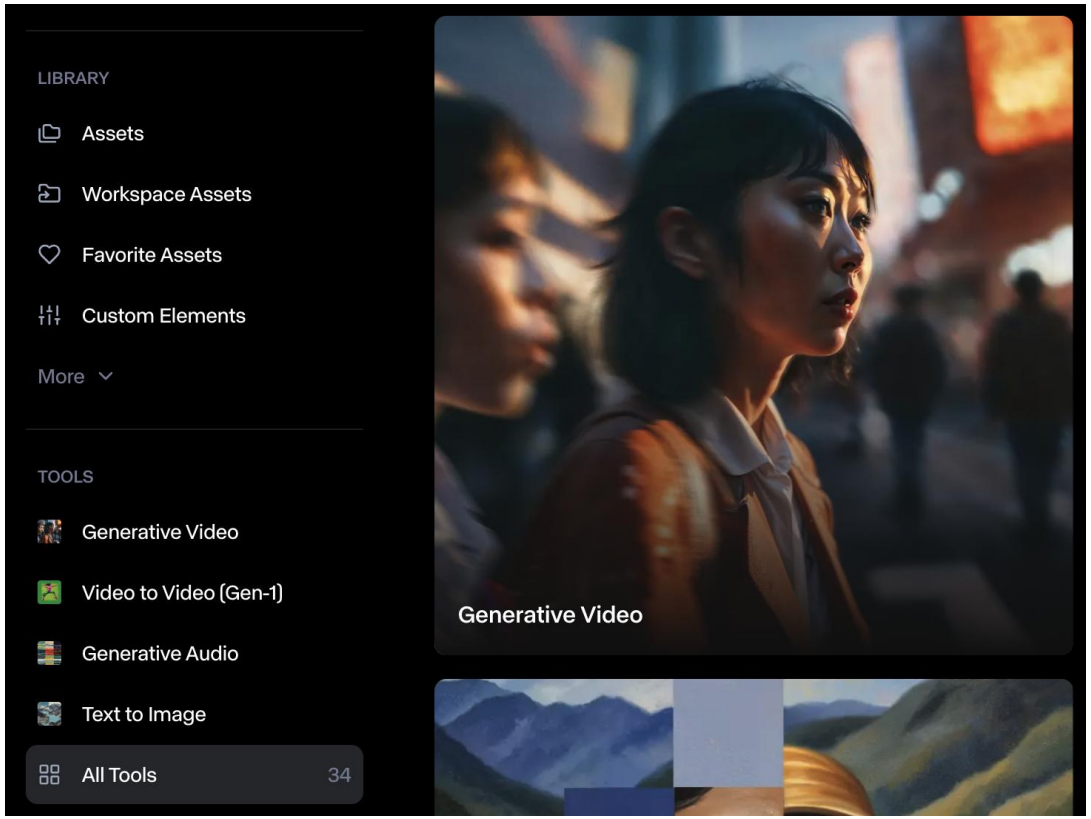


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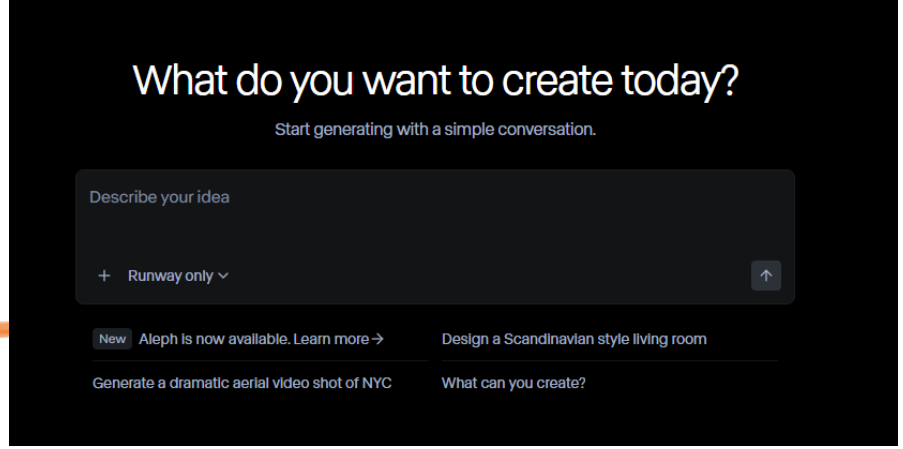
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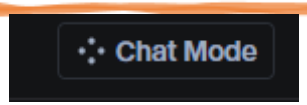
Text-to-video Generation

- A step-by-step guide to Text-to-Video Generation using Chat-Mode:
  - Select "Generative Video" from your dashboard.
  - Choose “Chat-mode“ (this is the default)
  - Select “runway only” to have a fixed cost
  - Select an aspect ratio for your image/video
  - Enter a text prompt describing the image/video you want to create
  - Interact with the AI assistant iteratively until you are satisfied with the result
    - It will try to understand your instructions to generate images/videos

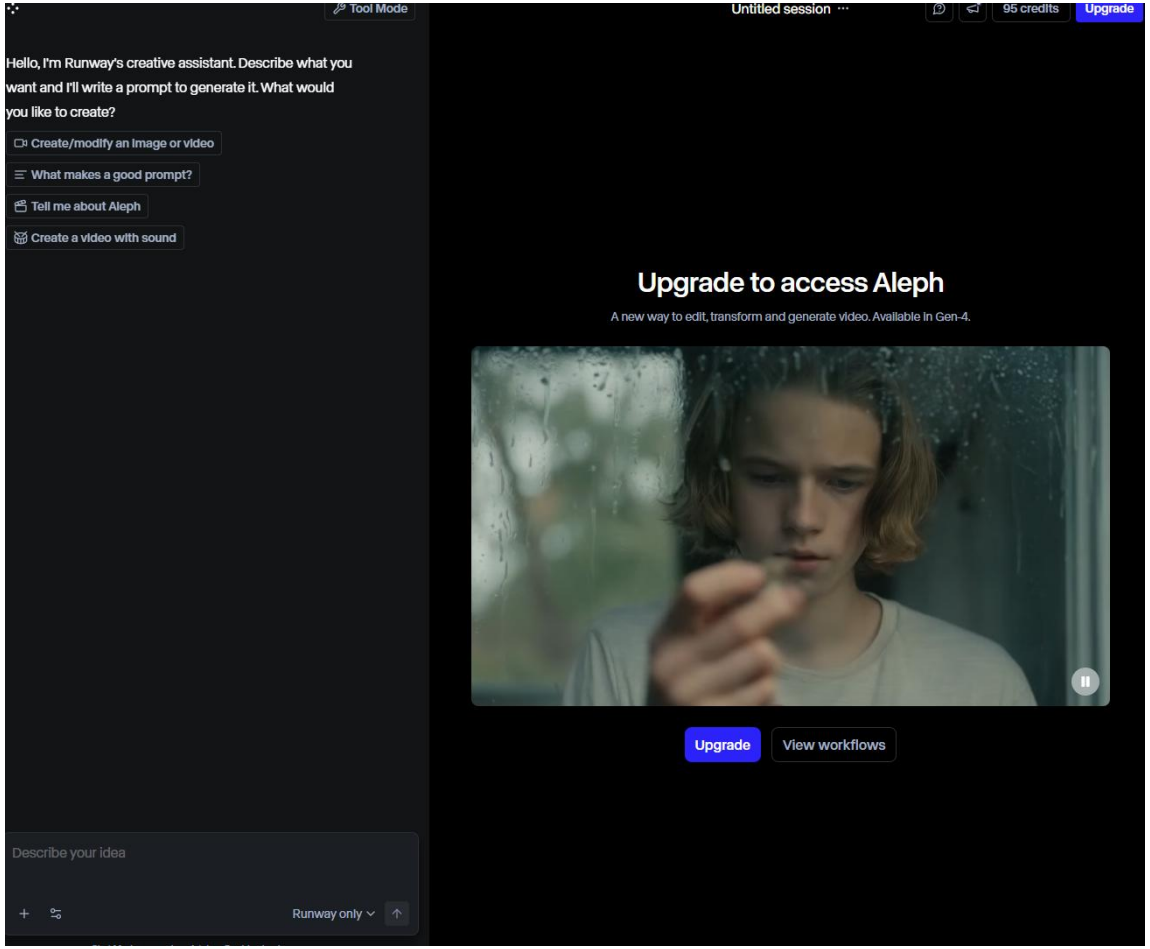


Text-to-video Generation



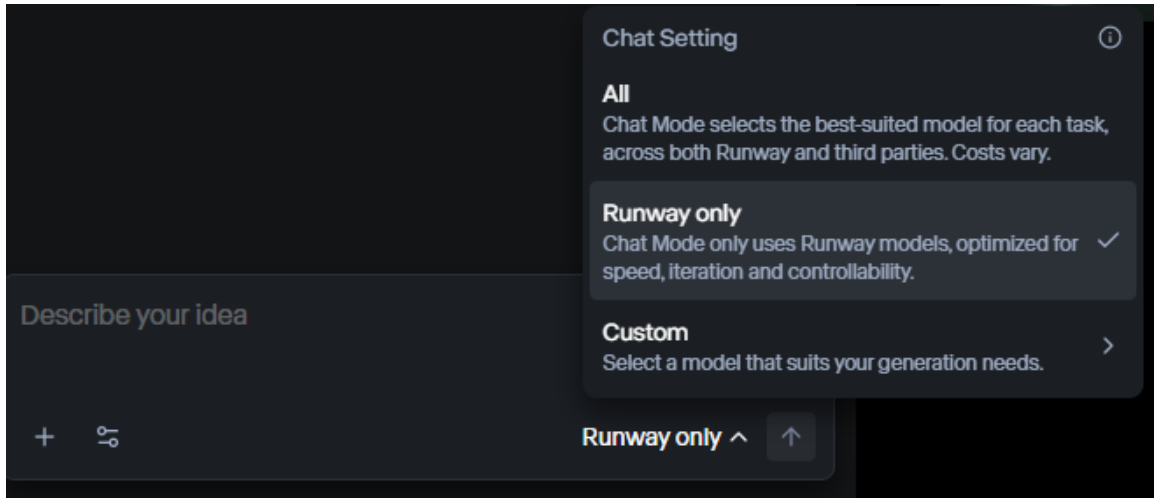


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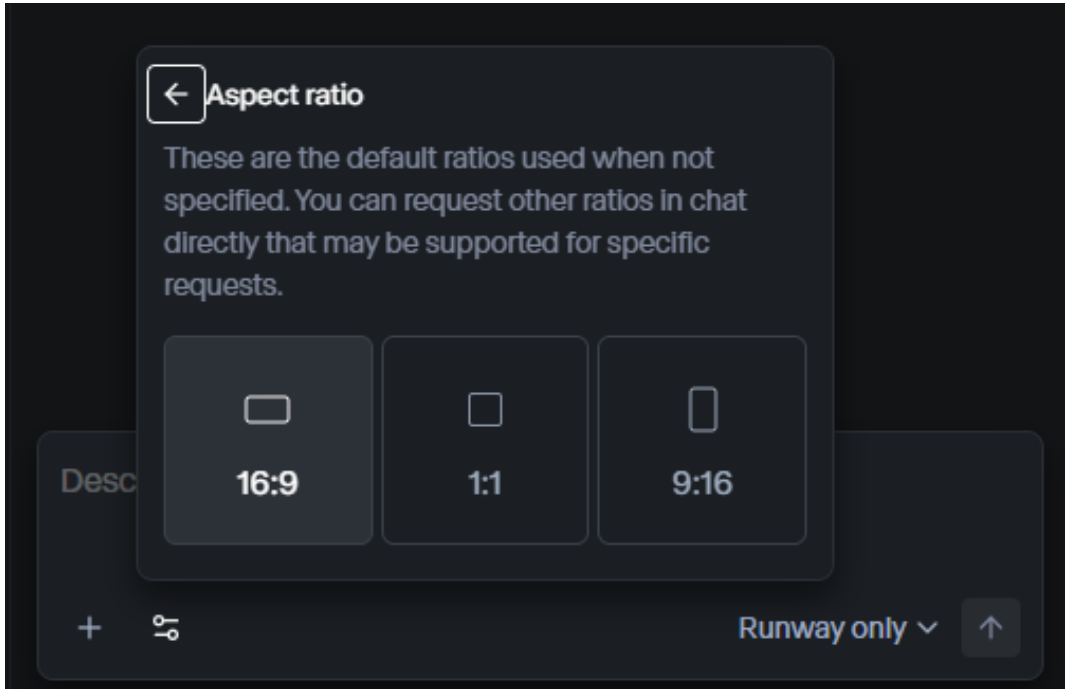
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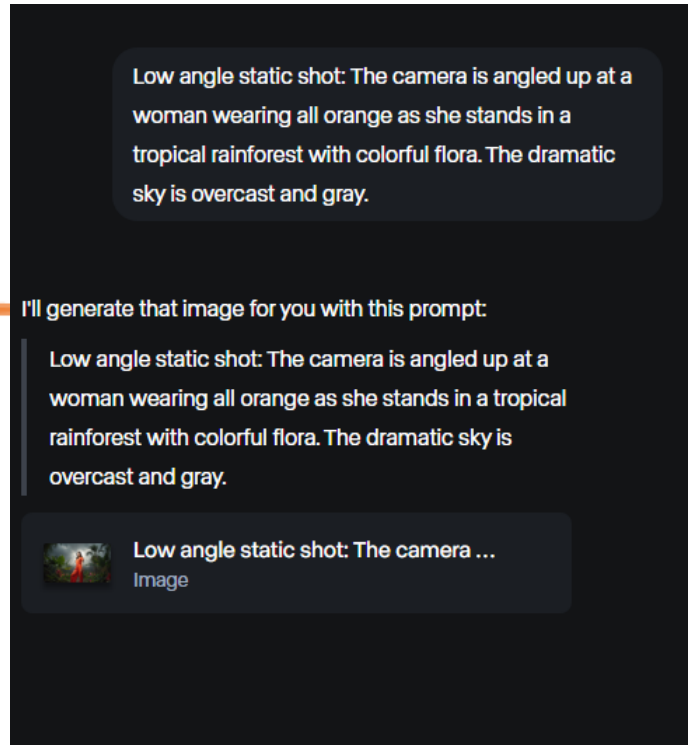
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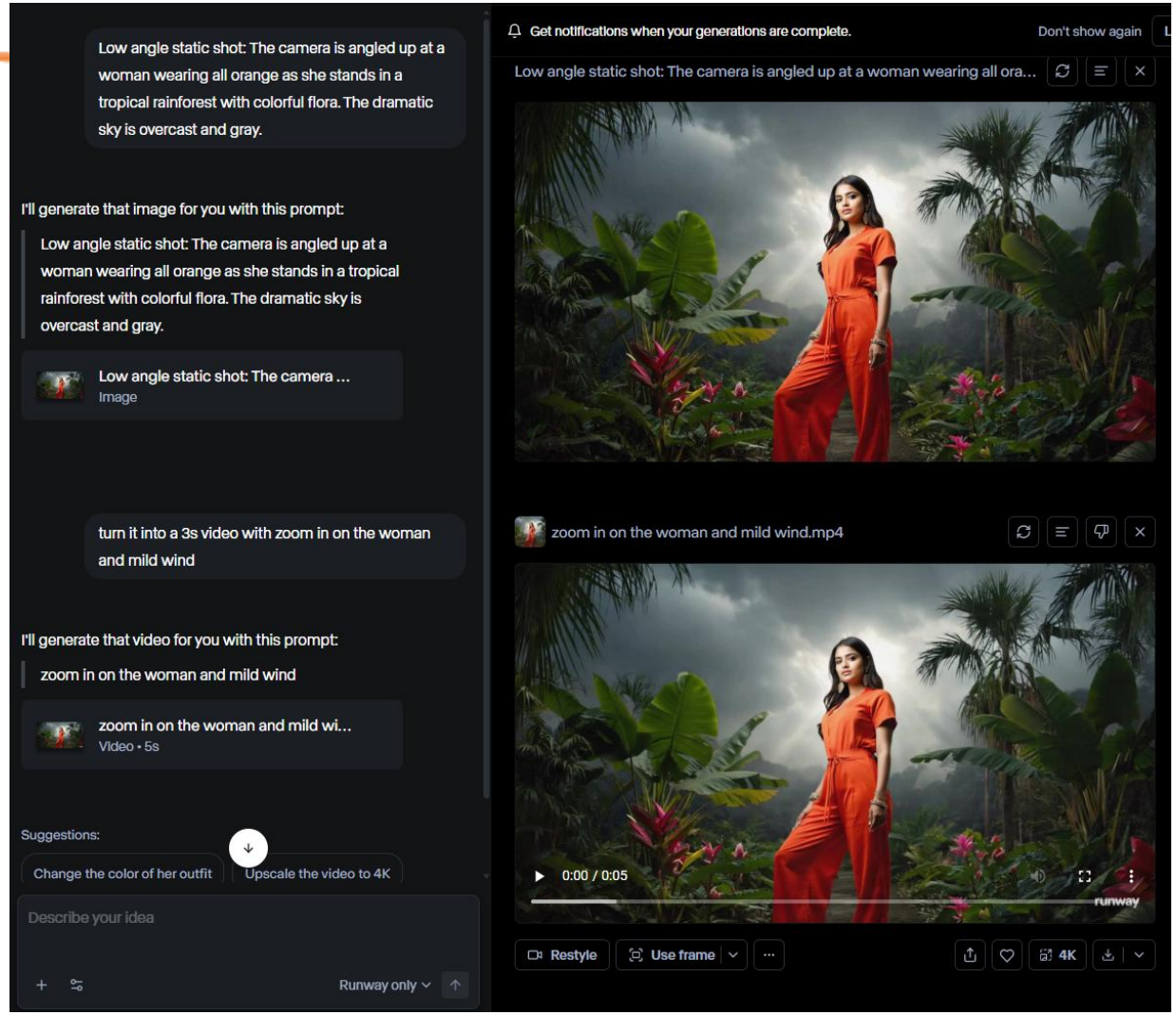
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  - Select an aspect ratio for your image/video
  - Enter a text prompt describing the image/video you want to create
    - E.g. Low angle static shot: The camera is angled up at a woman wearing all orange as she stands in a tropical rainforest with colorful flora. The dramatic sky is overcast and gray.
  - Interact with the AI assistant iteratively until you are satisfied with the result
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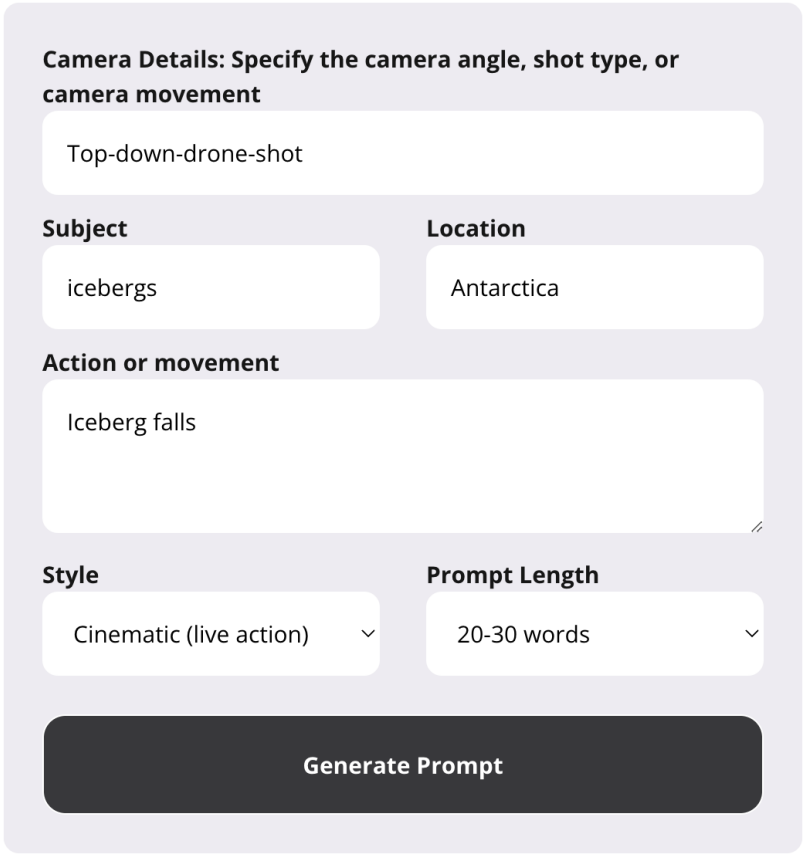
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  - Interact with the AI assistant iteratively until you are satisfied with the result
    - E.g. turn it into a 3s video with zoom in on the woman and mild wind
    - It will try to understand your instructions to generate images/videos



Text-to-video Generation

- What makes a good text prompt
  - A well-structured prompt typically follows this format: [camera movement]: [establishing scene]. [additional details].
    - Example: Top-down-drone-shot: a colossal iceberg in Antarctica collapses into the frigid sea. White and blue hues contrast under the piercing sunlight, creating a mesmerizing visual spectacle.
- How to generate a good text prompt
  - Navigate to the Runway Prompt Builder at https://word.studio/tool/runway-prompt-builder/
  - Set camera details: Low angle, High angle, Overhead, FPV (First Person View), etc. (refer to the following link)
  - Define the subject and location of your scene.
  - Describe the action or movement: Dynamic motion, Slow motion, Fast motion, etc. (refer to the following link)
  - Select a style for your video.
  - Choose the desired prompt length.
  - Click "Generate Prompt" to create your optimized text-to-video prompt.



https://help.runwayml.com/hc/en-us/articles/30586818553107-Gen-3-Alpha-Prompting-Guide

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  - Select Model and Duration
  - Select the image for generating video
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    - Upload an image
  - Provide additional text prompts alongside your image to better guide the video result.
    - Describe any movements you want to occur during the video. For example: "The elephant walks slowly in the desert..“
    - Describe new elements to appear throughout the video. For example: “Another elephant appears."
  - Click the "Generate" button. The model will interpret your image and prompts to produce the best possible video results.

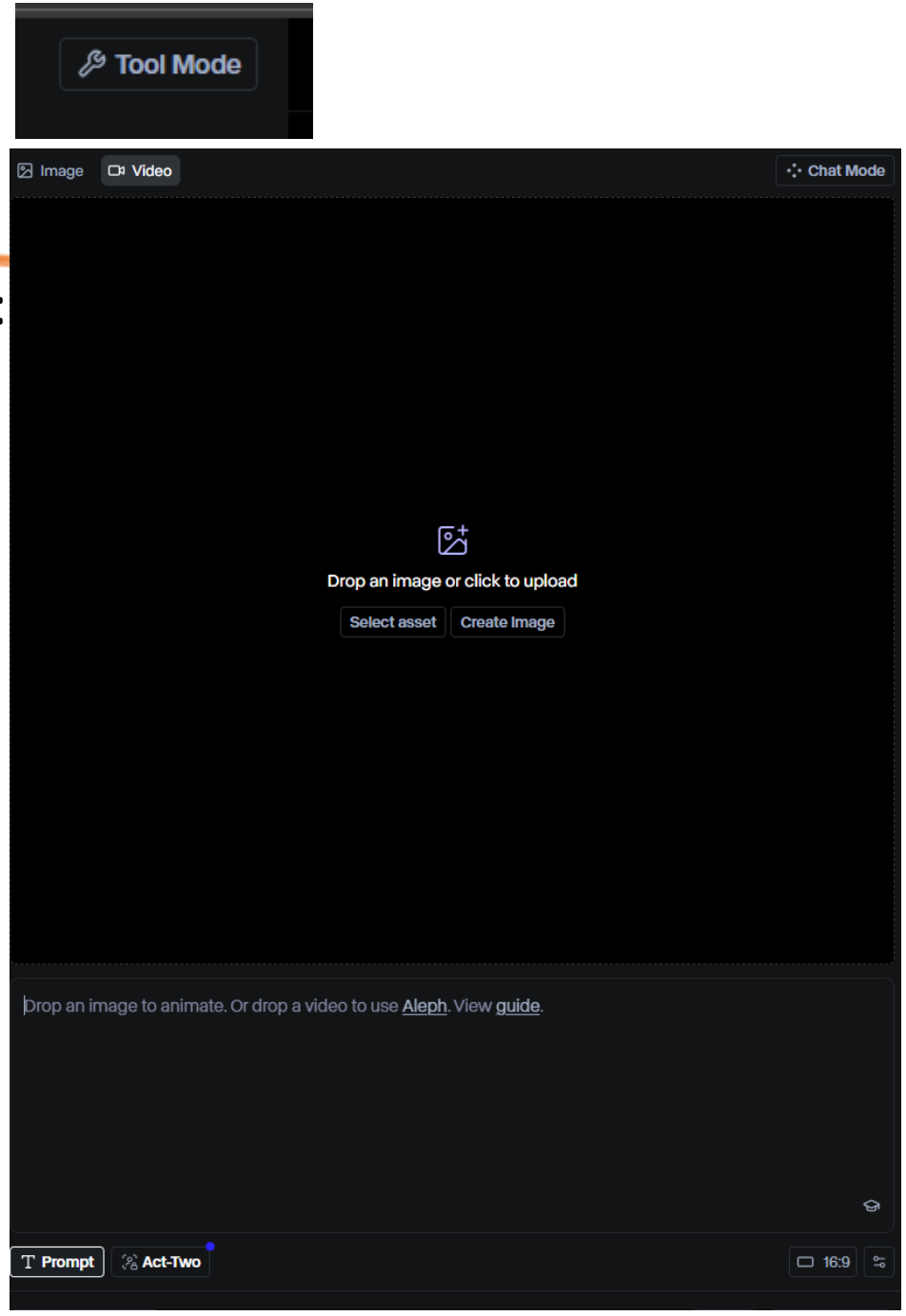


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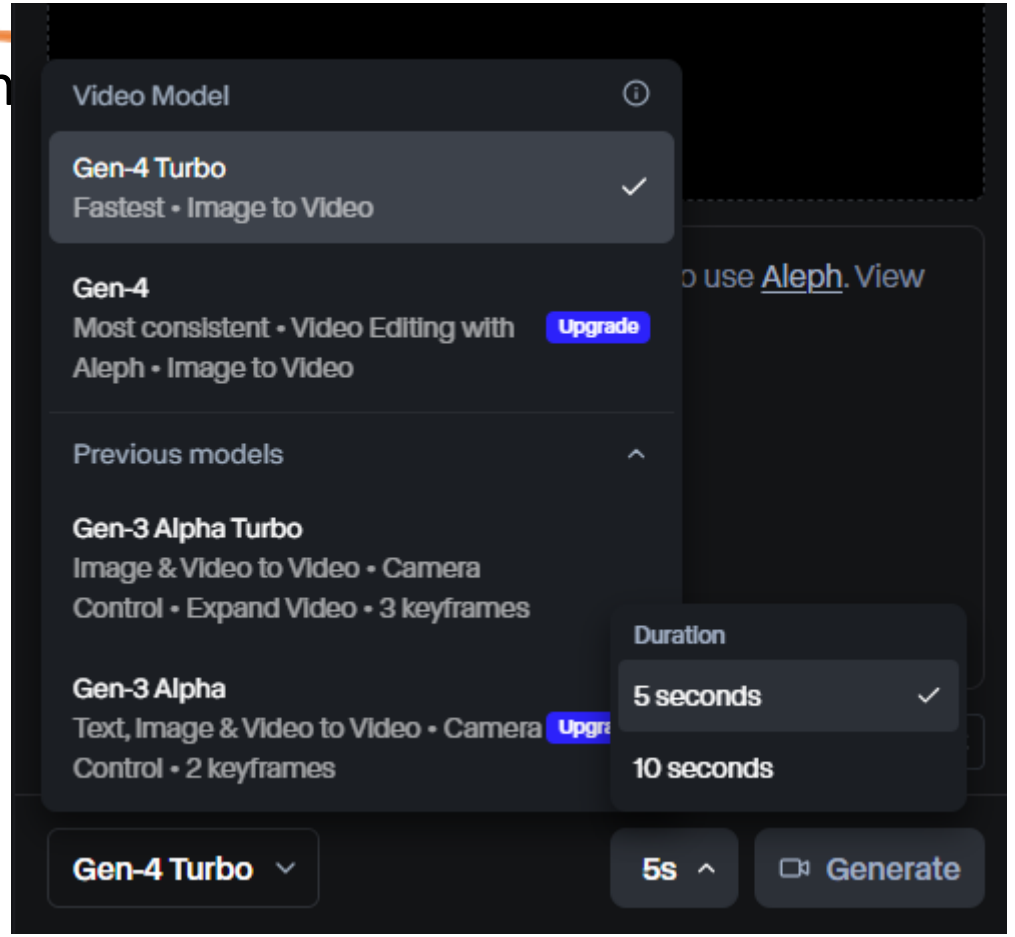


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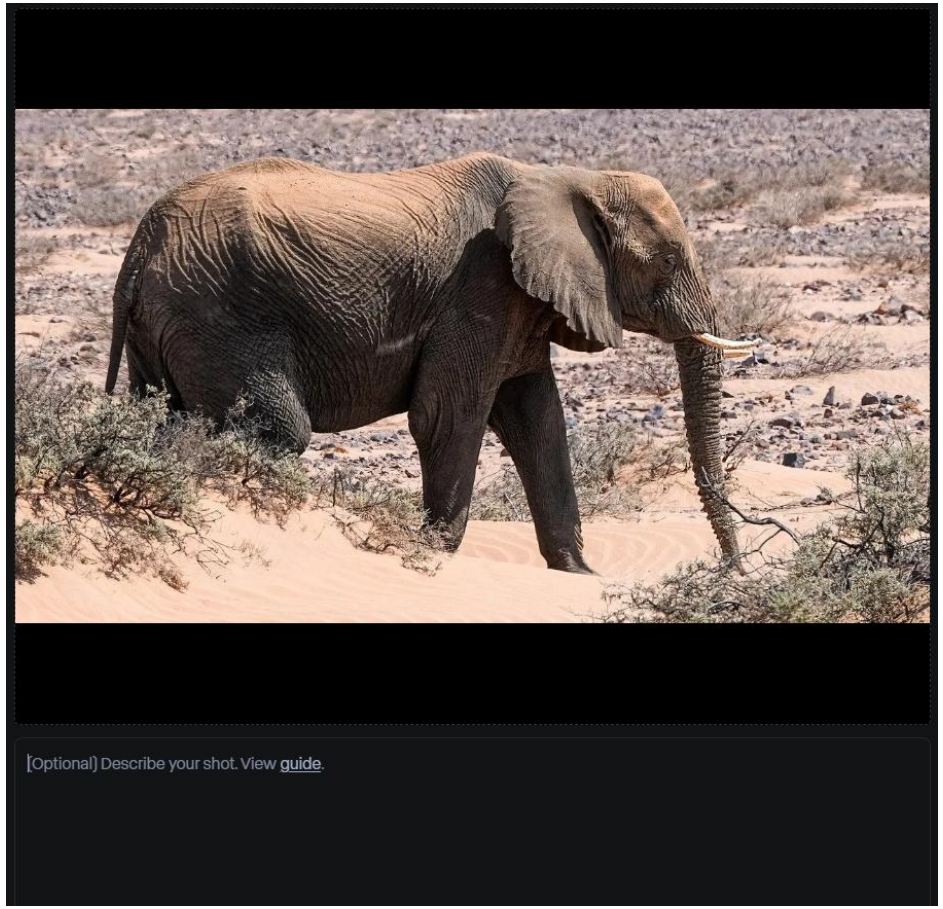


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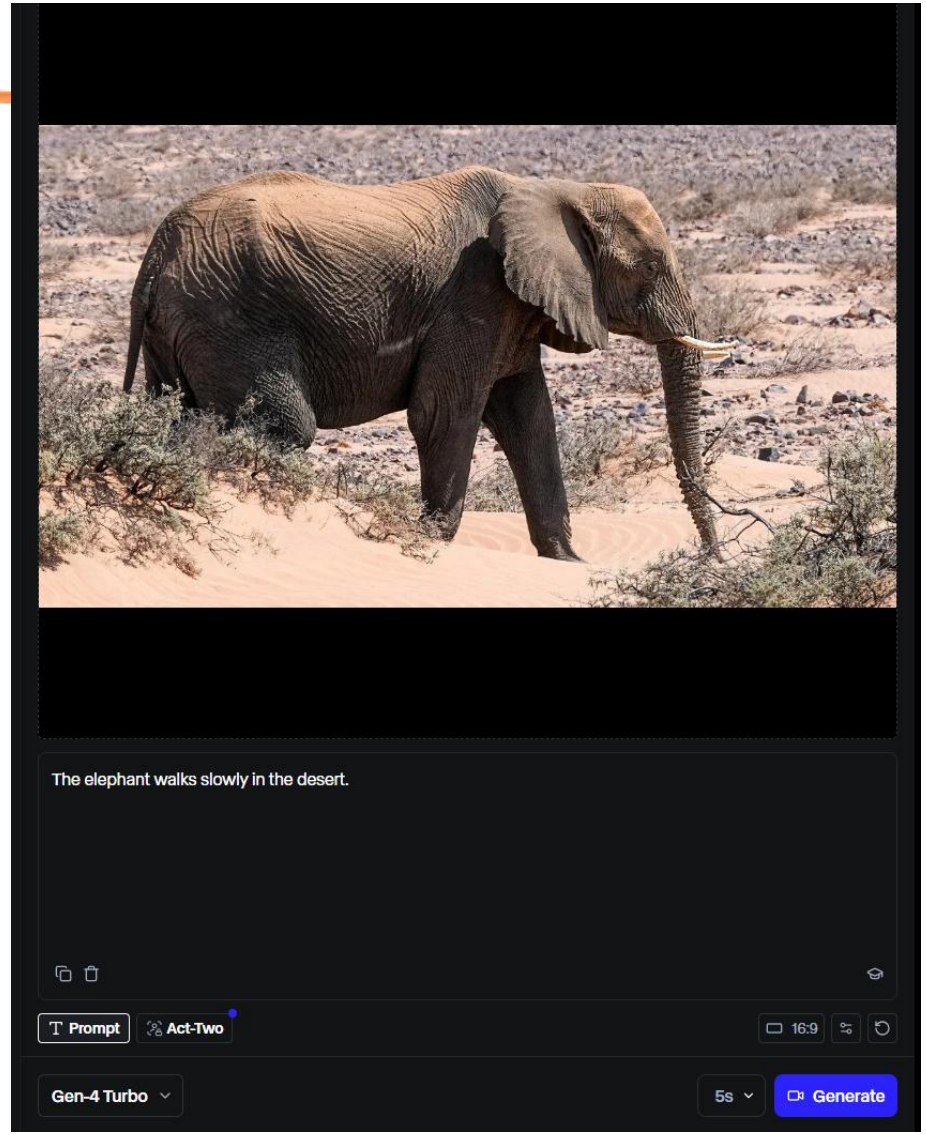
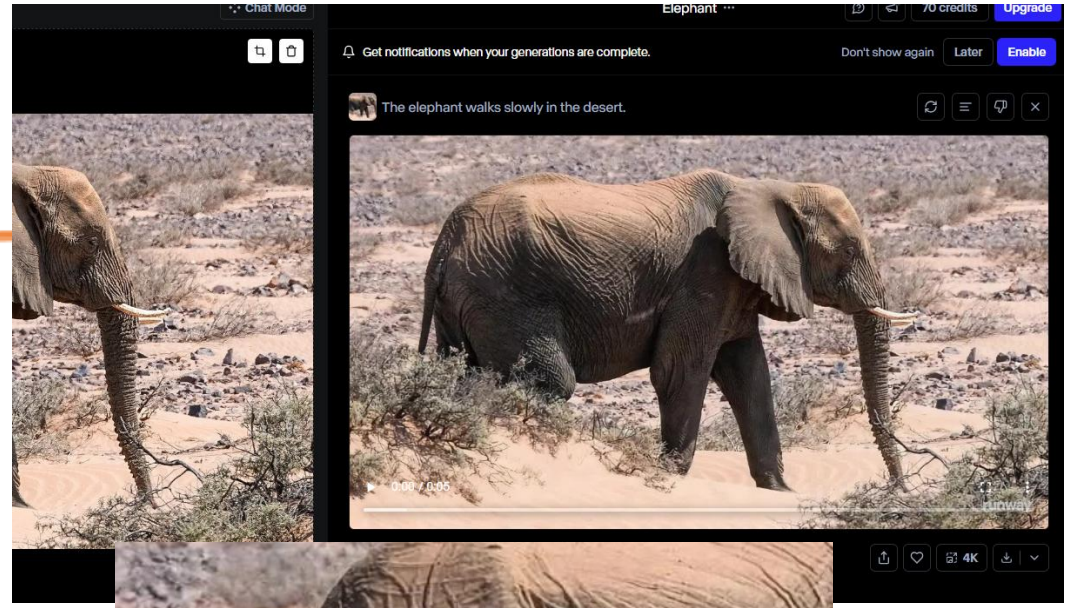
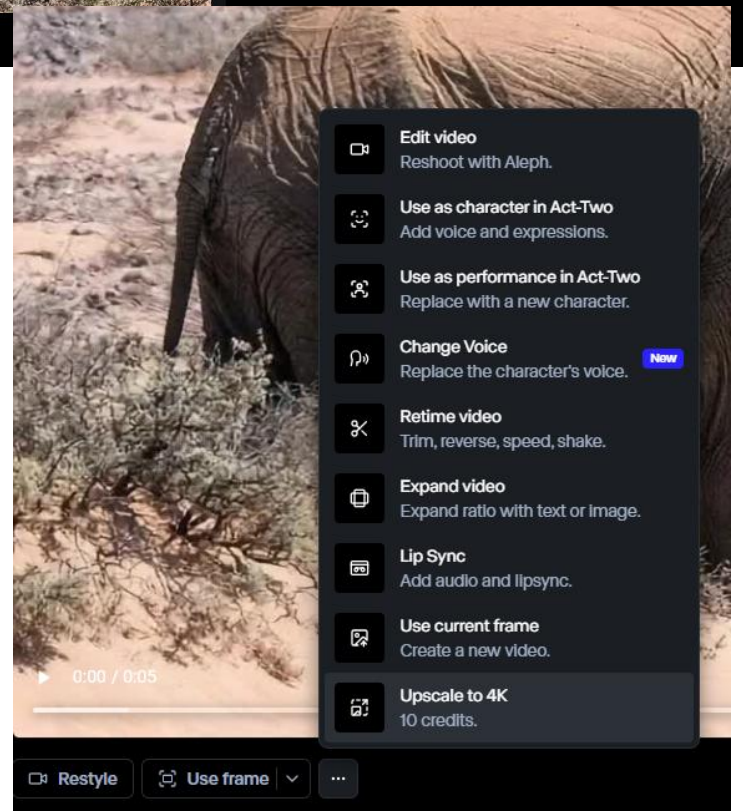


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  - Click the "Generate" button. The model will interpret your image and prompts to produce the best possible video results.
    - We can do further retouching after the generated video





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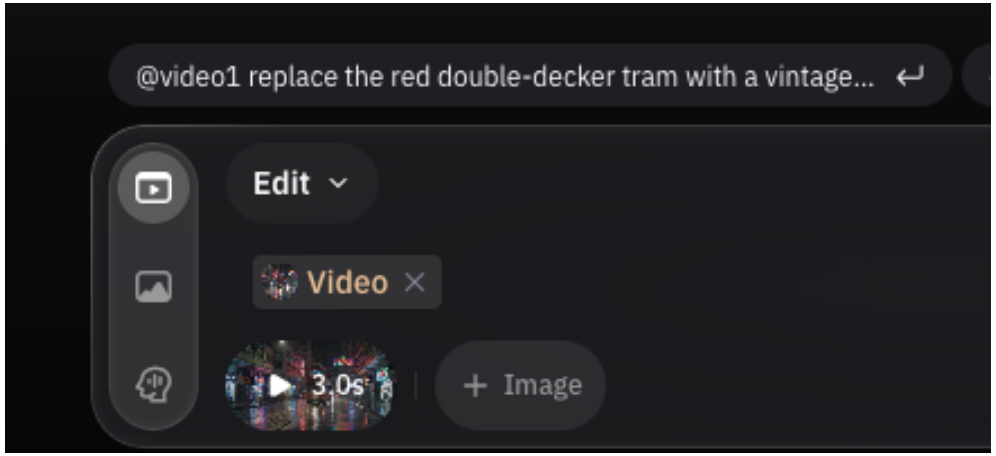
Video Editing

- A step-by-step guide to Video Editing on Wan:
  - Click "Video Edit" below the input box
  - Drag your video into the input box, or click "+ Video" to upload it.
  - Choose a resolution and duration
  - Enter a prompt describing what to change
  - Click ✦ to generate, and wait for the generation to finish



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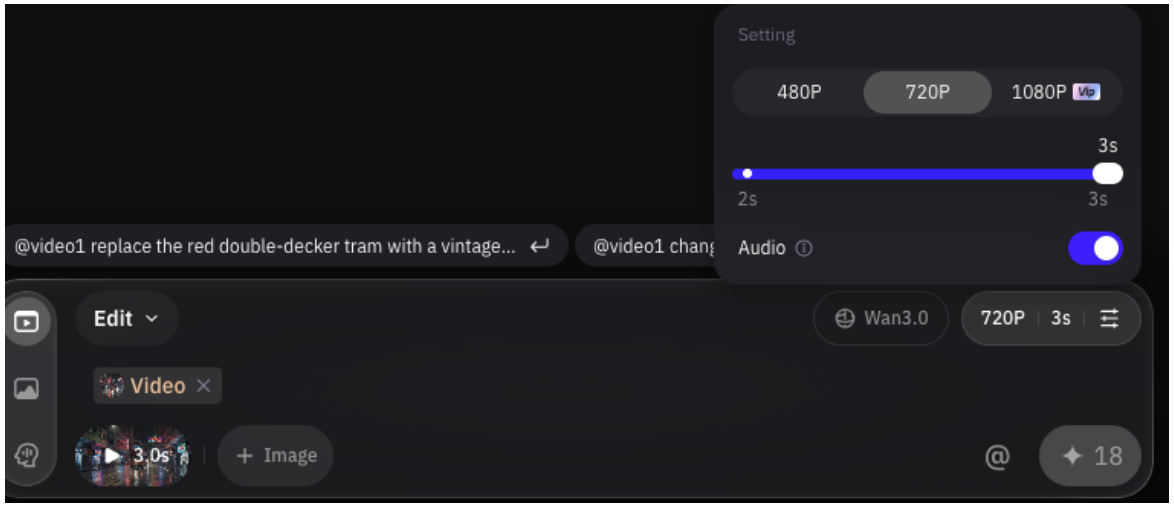
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The video appears as @video1 in the prompt.

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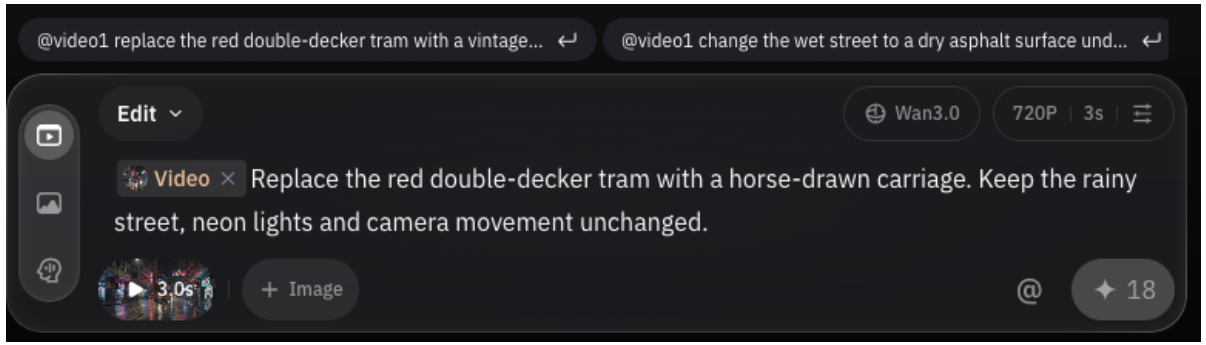
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  - Click "Video Edit" below the input box
  - Drag your video into the input box, or click "+ Video" to upload it.
  - Choose a resolution and duration
  - Enter a prompt describing what to change
  - Click ✦ to generate, and wait for the generation to finish



The duration follows the original video automatically (3 s here).

Video Editing

- A step-by-step guide to Video Editing on Wan:
  - Click "Video Edit" below the input box
  - Drag your video into the input box, or click "+ Video" to upload it.
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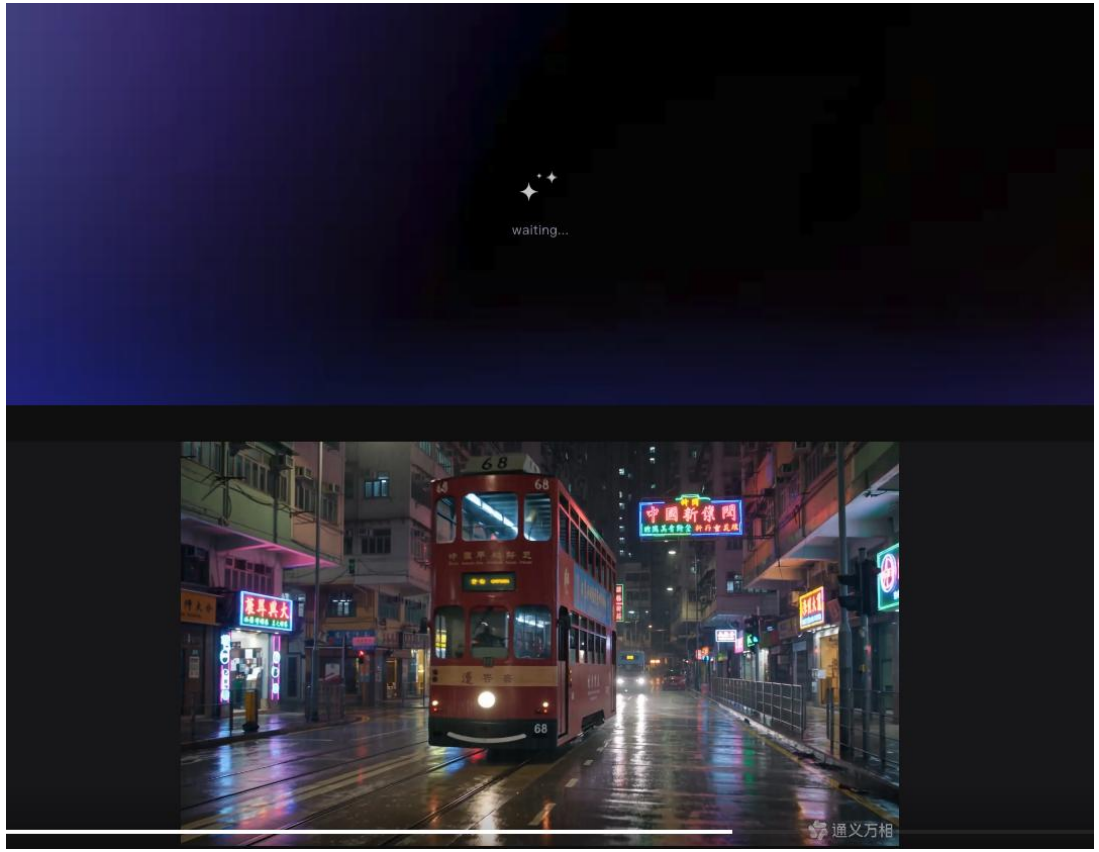


@video1 Replace the red double-decker tram with a horse-drawn carriage. Keep the rainy street, neon lights and camera movement unchanged.

Original prompt:
A red double-decker tram glides through a rainy Hong Kong street at night, neon signs reflecting on the wet road. Slow dolly-in camera, cinematic lighting. Sound: rain and a tram bell ringing.

Video Editing

- A step-by-step guide to Video Editing on Wan:
  - Click "Video Edit" below the input box
  - Drag your video into the input box, or click "+ Video" to upload it.
  - Choose a resolution and duration
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AI生成
Before
68
中國新傑閃
通义万相

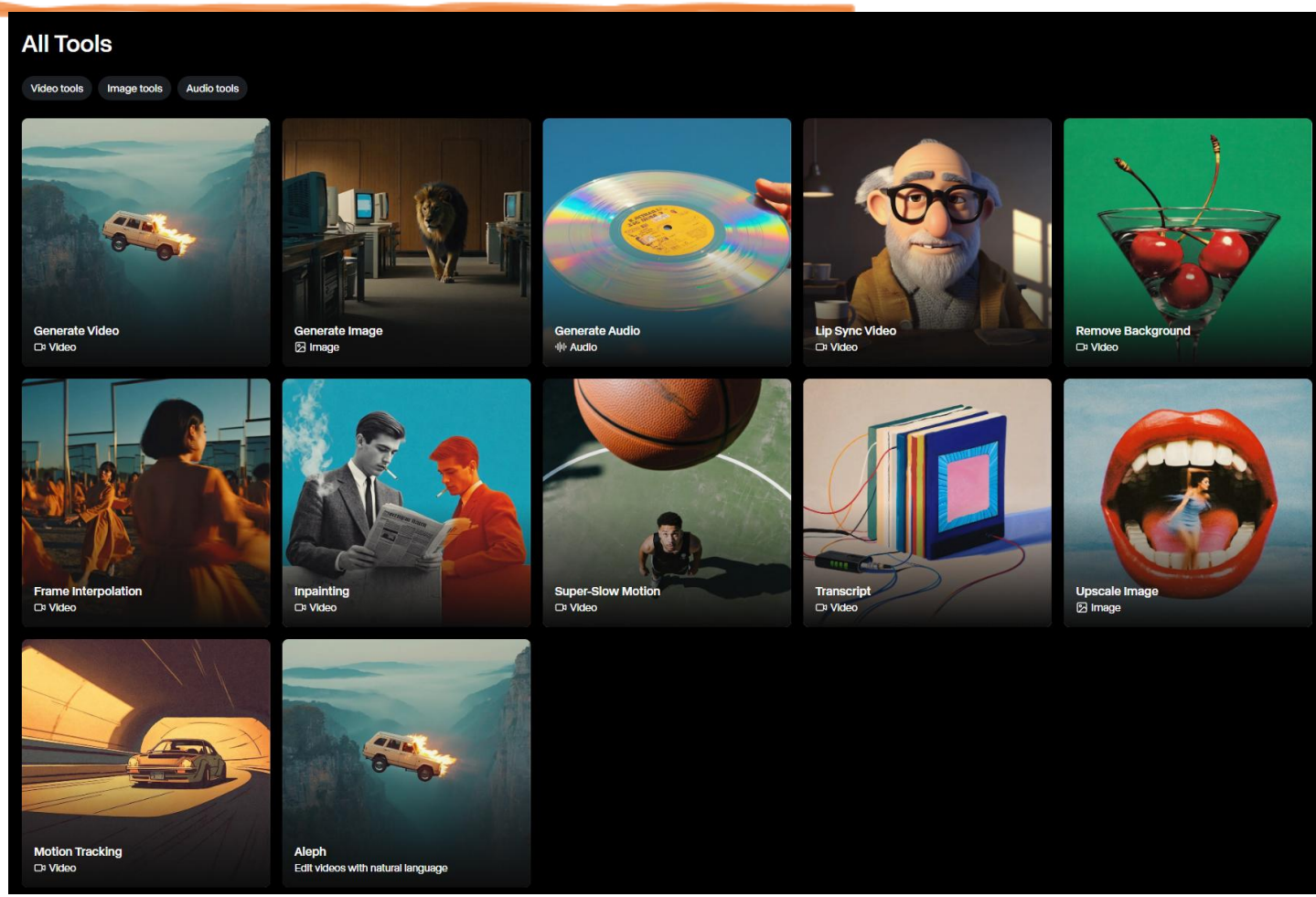

After
通义万相

Lab 04 Agenda

- Video Generation Models
  - Introductions of Video Generation Models
  - Products of Video Generation Models
  - Access to Video Generation Models
- Basic Video Generation Techniques
  - Text-to-video Generation
  - Image-to-video Generation
  - Video Edition
- Other Tools
  - Lip Sync Video
  - FramePack: Image-to-video Running Locally

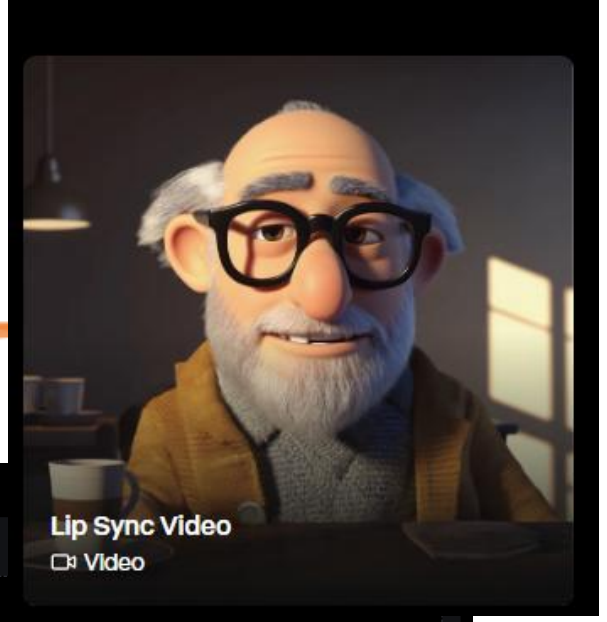
Other Tools

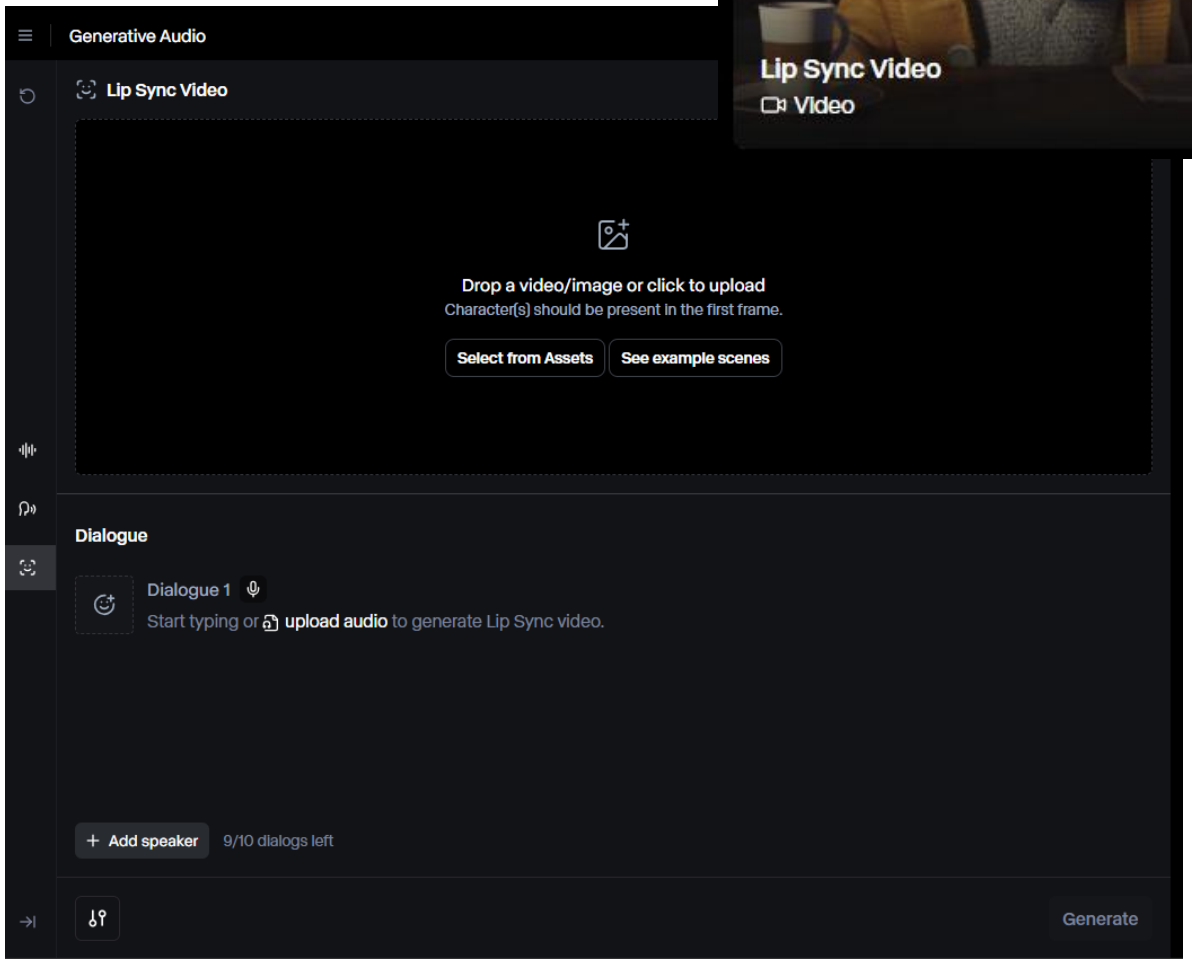
- Apart from the standard text-to-video, text-to-image, image-to-video, there are other tools available in runway
- Click “All Tools” will show up a bunch of GenAI video tools



Other Tools : Lip Sync Video

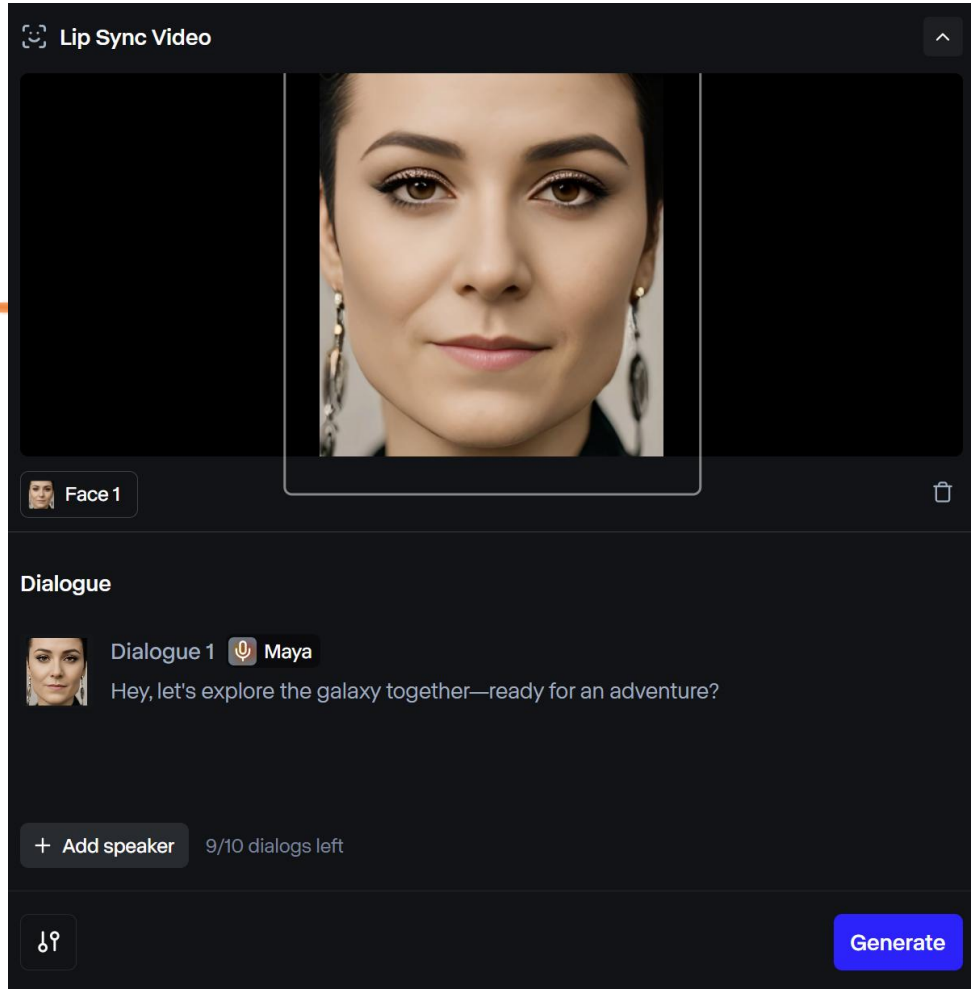
- The lip sync feature synchronizes text scripts or audio with character lip movements, delivering realistic dialogue in AI videos.
- Users can input audio or text, and Runway AI generates precise lip movements that match the spoken content.





Other Tools : Lip Sync Video

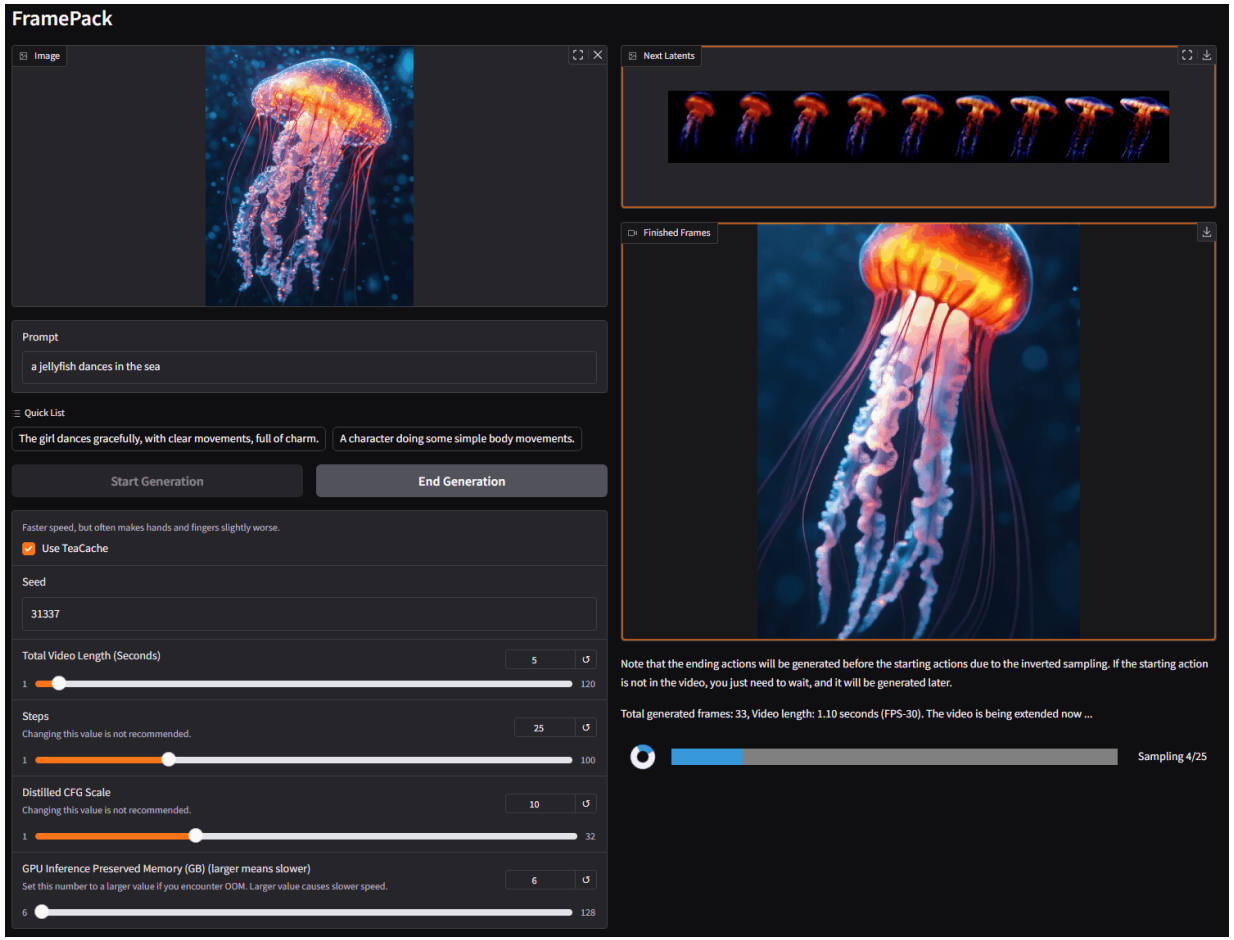
- Upload a face image
- Input the dialogue or speech of the character
  - E.g. Hey, let's explore the galaxy together—ready for an adventure?
- Click generate to obtain the lip synchronized video!





Framepack : Image-to-video running locally

- Visit project page of framepack https://lllyasviel.github.io/frame_pack_gitpage/
- Or directly download the windows package below: https://github.com/lllyasviel/FramePack/releases/download/windows/framepack_cu126_torch26.7z
- Depends on your PC's setting, it can take minutes or hours to finish



Question Time!

Or See you next week!